Pengzhen Li

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Final-year Ph.D. candidate in Economics specializing in applied econometrics, causal inference, and data-driven modeling for high-stakes decision-making. Experienced in building scalable, reproducible analytics in Python/R/SQL and translating large administrative/trade/production datasets into actionable insights for multidisciplinary stakeholders. Strong background in scenario/sensitivity analysis and counterfactual evaluation, with complementary experience in machine learning and time-series/panel methods.

Education

Ph.D. in Economics **Jan 2020 – Present**

University of Tennessee, Knoxville

M.A. in Finance **Sep 2016 – Jun 2018**

Shandong University, P.R. China

Thesis: The impact of renewable energy technological innovations on China's carbon dioxide emissions: Does the level of development matter? (Published in an ABS 3* journal, *Technological Forecasting and Social Change*; see paper. Cited 952 times; see Google Scholar.)

B.A. in Economics **Sep 2011 – Jun 2015**

Tongji University, Shanghai, P.R. China

Technical Skills

CFA Level II, IMPLAN, STATA, R, Python, GAMS, MATLAB, SQL, DAG, ArcGIS, Origin, etc.

Teaching

Primary Instructor

ECON 351 – Monetary Economics **Spring 2025**

ECON 322 – The Global Economy: Trade and Development **Summer 2025**

Teaching Assistant

ECON 511 – Microeconomic Theory **Fall 2023 – 2025**

ECON 581 – Mathematical Methods in Economics **Summer 2024**

ECON 512 – Advanced Microeconomics **Spring 2024**

AREC 525 – Agribusiness Operations Research Methods **Spring 2022 – 2023**

Research Projects

Graduate Research Assistant **Jan 2020 – Aug 2023**

The research focused on two projects funded by the National Institute of Food and Agriculture

(NIFA), USDA, and the FAA Office of Environment and Energy (ASCENT Project).

- Design cost-efficient or profit-maximizing forest-residue-based biofuel / sustainable aviation fuel (SAF) supply-chain networks using mixed-integer programming across the Southeast U.S.
- Manage high-resolution spatial data using ArcGIS, Origin, R, and Python.
- Conduct economic analysis (facility siting and investment decisions) using econometric models and the IMPLAN platform.
- Perform sensitivity analysis of feedstock quality and market conditions using extended mathematical programming (EMP) in GAMS, supervised machine learning, and @RISK.
- Conduct efficiency analysis for new preprocessing/conversion facilities or airports using Data Envelopment Analysis (DEA).

Other Research Experience

Machine learning

- Economic analysis of driving factors behind carbon dioxide emissions: evidence from patent data (supervised machine learning).
- Equity-premium prediction using *New York Times* and *Wall Street Journal* textual data (supervised and unsupervised machine learning).

Time series models

- Regime switching in the links between commodity prices using threshold VAR (TVAR).
- Volatility linkages and co-movements in wholesale commodity prices using VECM and multivariate GARCH (mGARCH).

Panel data models

- Effects of green technology innovation on carbon dioxide emissions across income levels using Hansen (1999) panel threshold models.
- Impacts of MFP and CFAP payments on U.S. non-real estate agricultural loans using GMM.
- Long-term impact of the 2008 Wenchuan earthquake on economic output across sectors/income groups in Sichuan using synthetic control methods (SCM).
- Multiple mediation analysis of digital finance and carbon dioxide emissions for Chinese listed firms using structural equation modeling (SEM).

Publications

Peer-Reviewed Publications

[1] **"Spatial optimization of the sustainable aviation fuel supply chains from forest residues via fast pyrolysis/hydrotreatment considering feedstock ash content variability."** (with T. Yu et al.), *Biomass and Bioenergy*, **208** (2026), 108793.

[2] **"Assessing the impact of preprocessing and conversion technologies on the sustainable aviation fuel supply from forest residues in the Southeast USA."** (with T. Yu et al.), *Transportation Research Record*, **2678**(9) (2024), 639–654.

[3] **“Beef price spread relationship with processing capacity utilization.”** (with C. Martinez et al.), *Journal of the Agricultural and Applied Economics Association*, **2**(1) (2023), 133–145.

[4] **“Do green technology innovations contribute to carbon dioxide emission reduction? Empirical evidence from patent data.”** (with K. Du and Z. Yan), *Technological Forecasting and Social Change*, **146** (2019), 297–303.

Working Papers

[1] **“Trade Restrictiveness Indices with Discriminatory Tariffs.”** (with James Lake).

[2] **“The Impact of Mayoral Endorsements on Presidential Election Outcomes.”** (with Matt Van Essen and Luiz Lima), *2024 J. Fred and Wilma Holly Award (best 2nd-year paper).*

[3] **Sustainable Aviation Fuel Production from Forest Residues with Ash Content: A Case Study of Nashville Airport.** (with T. Yu et al.).

[4] **Stochastic Optimisation of Logging Residue–based Biofuel Supply Chains Considering Feedstock Ash Content and Biofuel Price.** (with T. Yu et al.).

Work in Progress

- **Cooperative Multilateral Tariff Negotiations: A Quantitative Analysis Using Shapley Value.** (with Georg Schaur).

Presentations

- **Trade Restrictiveness Indices with Discriminatory Tariffs**, *Midwest Economic Theory and International Trade Meetings*, Virginia Tech, Blacksburg, Virginia, Spring 2025.
- **Assessing the Impact of Preprocessing and Conversion Technologies on SAF Supply Chains**, *Western Regional Science Association (WRSA) Annual Meeting*, Hawaii, 2023.
- **Economics of Sustainable Aviation Fuel from Forest Residues**, *Western Agricultural Economics Association (WAEA) Annual Meeting*, Santa Fe, New Mexico, 2022.
- **Impact of Harvesting Schemes on Woody Biomass Supply Chains**, *Southern Agricultural Economics Association (SAEA) Annual Meeting*, New Orleans, Louisiana, 2022.

Extra-Curricular Activities

Volunteer Tutor (Pro Bono)

Provided free tutoring for children from migrant families in Shanghai (including left-behind children), offering academic support and mentorship. Received the Outstanding Volunteer recognition.

Class Sports Representative

Organized and participated in multiple intra-college sports competitions, coordinating class participation and team logistics.

Interests

Table tennis (member, University of Tennessee Table Tennis Club); badminton; fishing; swimming; dog walking.